TECH6 H26: Mandatory Assignment 2

Submission and requirements

- Submit your solution individually on Canvas by **Thursday, November 5, 2026, at 12:00**.
- The assignment is graded **pass/fail**.
- You may use AI for support, but you may not submit output produced directly by AI as your own solution.
- Submit a single, executed Jupyter notebook that runs without errors after restarting the kernel and running all cells. Use the **TECH6** conda environment.
- Your notebook must read the data from the provided CSV file located in the same folder as your notebook using a relative path, i.e.,

```
import pandas as pd
df = pd.read_csv('lending_club.csv')
```

- Structure your notebook with clear sections and a logical flow. Define each function in its own code cell, and put each figure in its own code cell. A figure may contain several panels.
- Make figures readable, with axis labels, units, legends, and titles where appropriate. Keep displayed output concise and resolve execution warnings.
- Write reusable, well-formatted code. Avoid duplicating functions, add comments to explain nonobvious steps, and use efficient NumPy or pandas operations where appropriate.
- Explain and justify your choices and results in concise Markdown text, rather than in code comments.

Goal and common instructions

Your goal is to find the best-performing model for predicting loan interest rates among ordinary least squares (OLS), Ridge, and Lasso. You choose the predictors, transformations, treatment of categorical variables, preprocessing, and model complexity.

- Use a seed of 5678 for every operation involving a random number generator (`random_state=5678`).
- Use cross-validation (CV) for Ridge and Lasso model choices. CV is optional for OLS. Tune multiple hyperparameters jointly, where applicable.
- Use a common sample to compare models and the same folds whenever you use CV.
- Use root mean squared error (RMSE) as the scoring metric for cross-validation.
- Reserve the test sample for final evaluation in Part 6.
- Report model performance on the test sample using both RMSE and R^2.

Data

The file `lending_club.csv` contains 10,000 loans issued through Lending Club, a US lending platform, between January and March 2018. It is a teaching extract of OpenIntro's dataset.

Each row describes a loan and the applicant's financial situation. The outcome is `interest_rate`, measured in percent per year.

Variable	Description
`interest_rate`	Annual loan interest rate, in percent; the outcome.
`loan_amount`	Amount borrowed, in USD.
`term`	Loan duration: 36 or 60 months.
`annual_income`	Applicant's annual income, in USD.
`debt_to_income`	Monthly debt payments as a percentage of monthly income.
`emp_length`	Completed years in the current job, capped at 10.
`homeownership`	Housing status: `RENT`, `MORTGAGE`, or `OWN`.
`verified_income`	Verification: `Not Verified`, `Source Verified`, or `Verified`.
`application_type`	Individual or joint application.
`loan_purpose`	Stated purpose of the loan.
`credit_history_years`	Loan issue year minus the year of the first credit line.
`inquiries_last_12m`	Credit inquiries in the preceding 12 months.
`open_credit_lines`	Number of open credit accounts.
`total_credit_limit`	Existing credit limit excluding mortgages, in USD.
`credit_utilization`	Existing balance divided by limit, excluding mortgages.
`delinq_2y`	Credit-line delinquencies in the preceding two years.
`bankruptcy`	Number of bankruptcies in the applicant's public record.

`credit_utilization` is a fraction: 0.5 means 50%. Interest rate and debt-to-income are percentages: values of 12 and 25 mean 12% and 25%, respectively. Utilization can exceed 1. Credit limits and balances describe existing accounts across lenders, excluding mortgages; we interpret them as the applicant's credit position at application.

A credit line is an account through which someone can borrow. An inquiry occurs when a lender checks an applicant's credit history. A delinquency is an overdue payment; it does not necessarily involve bankruptcy. `bankruptcy` counts bankruptcy records and has no specified two-year window.

The data include individual and joint applications. For joint applications, `annual_income` and `debt_to_income` describe the individual applicant, rather than the applicants' combined finances.

Part 1 — Prepare the data

1. Load `lending_club.csv` from the notebook's directory and drop any rows with missing values. After every step that removes observations, print the number dropped and the number remaining.
2. Create a homeownership indicator variable from the categorical `homeownership` column:

   `MORTGAGE` or `OWN` = 1; `RENT` = 0.
3. Create an income-verification indicator variable from the categorical `verified_income` column:

   `Source Verified` or `Verified` = 1; `Not Verified` = 0.
4. Decide which predictors to use in levels or natural logs. Keep the outcome, `interest_rate`, in levels. Discuss your choices, including how you will handle categorical variables and any nonpositive values that affect taking logs.
5. Split the common sample into 50% training and 50% test observations using. Keep the test sample for Part 6.

Part 2 — Explore the training data

1. Choose informative plots for selected candidate predictors and explain your choices. Match the plot type to the variable: for example, boxplots for discrete predictors and scatter plots of interest rates against continuous predictors, with binned means overlaid. You do not need to plot every predictor.
2. Tabulate the numeric predictors' correlations with the interest rate.
3. Discuss the patterns, unusual observations, and what these patterns suggest about predicting interest rates.

Use only the training data for this part.

Part 3 — Ordinary least squares

1. Develop an OLS model and explain your choices of predictors, transformations, treatment of categorical variables, and preprocessing.
2. Optionally, use cross-validation (CV) to assess the specification and select model hyperparameters. If you consider polynomial features, you only need to consider degrees 1–3 to keep computation manageable.
3. If you use CV, report CV performance and any selected hyperparameter values.
4. If applicable, plot validation curves to illustrate how model performance changes as you vary the hyperparameters.

Keep the test sample reserved for Part 6.

Part 4 — Ridge regression

1. Develop a Ridge model and choose its hyperparameters by CV. If you select more than one hyperparameter, tune them jointly.
2. Explain your specification and preprocessing choices.
3. Report CV performance and any selected hyperparameters.
4. Plot validation curves to illustrate how model performance changes as you vary the hyperparameters.
5. Check whether your search range adequately covers the best-performing penalties.

Keep the test sample reserved for Part 6.

Part 5 — Lasso

1. Develop a Lasso model and choose its hyperparameters by CV. If you select more than one hyperparameter, tune them jointly.
2. Explain your specification and preprocessing choices.
3. Report CV performance and any selected hyperparameters.
4. Plot validation curves to illustrate how model performance changes as you vary the hyperparameters.
5. Check whether your search range adequately covers the best-performing penalties.

Keep the test sample reserved for Part 6.

Part 6 — Compare models

1. Compare your candidate models and explain the overall choice using CV performance where available. Make this choice before inspecting test errors.
2. Report test RMSE and test R^2 for the final fitted candidates, together with their specifications, including the number of features and any selected hyperparameter values. Include CV performance where available.
3. Plot observed versus predicted test interest rates for your chosen model, including an equality line. Do the prediction errors exhibit any patterns?
4. The supplied data omit some information lenders use to set interest rates, such as borrowers' credit scores. Discuss how this missing information might explain the models' prediction errors.

Use the unchanged test sample for final evaluation. Do not revise model choices using test errors.

Part 7 — AI statement

Describe how you used AI in preparing your solution. Also state which vendor, model, subscription, and tools you used (e.g., Fable 5 with Claude Max 200 using Claude Code CLI).